

OTTAWA CDA, ON, Canada

WMO: 710630

Lat: 45.3825N

Lon: 75.7139W

Elev: 79

StdP: 100.38

Time Zone: -5.00 (NAE)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.6	-20.8	-28.6	0.3	-23.0	-25.4	0.4	-20.1	9.3	-6.3	8.5	-8.5	2.9	270	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	30.9	21.8	29.1	21.0	27.5	20.4	23.8	28.3	22.7	26.9	21.7	25.8	4.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	17.3	26.0	21.3	16.1	25.1	20.4	15.2	24.1	71.7	28.3	67.5	26.8	64.0	25.8	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.5	6.7	DB	-26.4	33.8	2.9	1.6	-28.5	35.0	-30.2	35.9	-31.8	36.8	-33.9	38.0	(4)
(5)			WB	-26.7	25.5	2.8	1.5	-28.7	26.5	-30.3	27.4	-31.9	28.2	-33.9	29.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	-9.3	-7.9	-2.2	6.2	13.7	18.6	21.3	20.3	16.1	8.8	2.1	-5.1	(6)	
	DBStd	11.81	6.84	5.85	6.04	4.95	4.36	3.62	2.92	3.00	4.02	4.50	4.84	6.02	(7)	
	HDD10.0	2412	598	500	381	133	14	0	0	0	2	78	239	468	(8)	
	HDD18.3	4404	857	733	637	364	154	41	7	14	90	295	487	726	(9)	
	CDD10.0	1306	0	0	2	20	129	258	349	319	186	42	2	0	(10)	
	CDD18.3	257	0	0	0	1	11	48	97	75	24	1	0	0	(11)	
	CDH23.3	2044	0	0	1	13	118	390	778	542	194	8	0	0	(12)	
	CDH26.7	534	0	0	0	2	27	102	223	137	43	1	0	0	(13)	
(14) Wind	WSAvg	3.2	3.4	3.6	3.7	3.8	3.4	2.9	2.7	2.6	2.6	3.1	3.3	3.4	(14)	
(15) Precipitation	PrecAvg	939	70	54	62	77	76	94	90	85	92	86	74	80	(15)	
	PrecMax	1182	124	104	127	156	127	192	196	163	160	182	111	136	(16)	
	PrecMin	793	30	23	23	14	27	17	18	38	44	22	23	26	(17)	
	PrecStd	93	25	21	28	34	27	39	42	24	33	34	24	29	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	6.1	17.5	25.2	29.6	32.0	33.2	32.9	30.7	24.3	16.7	10.2	(19)	
		MCWB	6.9	4.4	13.1	16.4	20.8	22.8	23.3	22.4	22.4	18.0	12.9	8.3	(20)	
	2%	DB	4.4	4.0	11.0	20.7	26.5	29.5	31.0	29.8	27.6	21.0	13.9	7.0	(21)	
		MCWB	3.3	2.0	7.1	12.9	18.0	21.3	21.9	21.6	20.9	16.4	11.4	5.8	(22)	
	5%	DB	2.4	2.5	7.6	17.3	24.0	27.3	29.1	27.9	25.5	18.4	11.1	4.3	(23)	
		MCWB	1.4	1.1	4.4	11.1	16.1	20.1	21.1	20.5	19.5	14.5	8.5	3.2	(24)	
	10%	DB	0.5	1.0	5.5	14.3	21.7	25.5	27.4	26.4	23.5	15.9	9.0	2.5	(25)	
		MCWB	-0.3	-0.1	2.9	9.2	15.0	19.0	20.5	20.0	18.7	12.7	6.6	1.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.0	4.9	13.4	17.5	22.1	24.7	25.7	24.7	23.4	20.0	14.1	8.5	(27)	
		MCDB	7.2	6.1	17.2	22.9	27.4	28.8	30.7	28.8	28.3	22.7	15.6	9.4	(28)	
	2%	WB	3.6	2.4	7.7	14.5	19.9	23.1	24.0	23.0	21.9	17.0	11.4	6.0	(29)	
		MCDB	4.2	3.5	10.5	18.9	24.6	27.3	28.5	27.0	26.2	20.3	13.4	7.2	(30)	
	5%	WB	1.4	1.2	4.9	11.9	18.0	21.6	22.8	22.1	20.6	15.0	8.9	3.3	(31)	
		MCDB	2.2	2.2	7.1	16.1	21.9	26.0	26.9	26.1	24.5	17.8	10.9	4.2	(32)	
	10%	WB	-0.2	0.0	3.2	9.7	16.1	20.2	21.7	21.1	19.2	13.2	7.0	1.6	(33)	
		MCDB	0.6	1.0	5.1	13.8	20.3	23.7	25.6	24.6	22.4	15.4	8.6	2.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	8.9	8.8	9.7	10.5	9.9	10.1	10.1	10.6	8.4	7.3	6.8	(35)	
		MCDBR	8.9	9.3	10.9	15.5	14.1	12.6	12.2	12.3	12.9	12.6	10.7	7.9	(36)	
	5% WB	MCWBR	8.6	8.3	7.6	9.1	7.4	6.2	5.6	5.6	6.9	8.1	8.4	7.1	(37)	
		MCWBR	8.6	8.8	10.1	13.6	11.8	10.8	10.3	10.3	11.6	11.6	10.2	7.8	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.264	0.288	0.325	0.346	0.382	0.415	0.419	0.401	0.371	0.340	0.310	0.279	(40)	
		taud	2.379	2.302	2.350	2.412	2.349	2.270	2.265	2.321	2.397	2.480	2.492	2.393	(41)	
	Ebn at Noon	Ebn at Noon	858	894	906	915	886	854	846	848	846	828	796	794	(42)	
		Edh at Noon	73	96	106	109	121	131	130	119	101	81	67	66	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.39	2.36	3.67	4.49	5.20	5.65	5.74	5.06	3.96	2.33	1.43	1.02	(44)	
	RadStd	RadStd	0.17	0.23	0.31	0.45	0.51	0.44	0.47	0.24	0.33	0.21	0.16	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47) Regional (32 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)	

Nomenclature: See separate page